

PROJECT SYNOPSIS

Proposed Computer Science / Software Development Project

Proposed Project Title	Web-Based Enterprise Resource Planning and Information Management System
Area	IT / Digital Infrastructure and Software Development
Proposed By	B.Tech CSE Student
Duration	As per the internship/project schedule

1. Introduction

The proposed project is a web-based information management system designed to organize routine records, requests, and work-related information in one place. In many organizations, information is maintained through different files, registers, emails, or separate software tools. This can make searching, updating, and tracking records time-consuming. The proposed system will provide a simple and structured platform for managing information and improving coordination between users and departments.

2. Problem Statement

When information is stored in separate places, it becomes difficult to maintain updated records and obtain the required details quickly. Repeated data entry, delayed communication, and limited tracking can also affect the efficiency of daily work. A centralized and user-friendly system is therefore needed to reduce these difficulties and make information access more organized.

3. Objectives

- To develop a centralized web-based platform for managing organizational information.
- To provide secure login and role-based access for different users.
- To reduce repeated manual work and make record searching easier.
- To allow users to add, update, view, and track relevant information.
- To maintain organized records that can be used for reporting and future reference.
- To design a system that can be extended according to the requirements of a particular department.

4. Proposed Features

- **User Login and Access Control:** Different users will be provided access according to their roles.
- **Centralized Records:** Important information will be stored in a structured database.
- **Record Management:** Authorized users can add, edit, view, and update records.
- **Search and Filtering:** Users can find required information without checking multiple files.
- **Status Tracking:** Requests or tasks can be tracked through different stages.
- **Reports and Summary:** Basic reports can be generated for monitoring and decision-making.
- **Responsive Interface:** The system will be designed to work on commonly used desktop and mobile screens.

5. Proposed Technology

Frontend: HTML, CSS, JavaScript and React.js

Backend: Node.js and Express.js

Database: MongoDB

Other Tools: Git/GitHub, REST APIs and suitable authentication methods

6. Expected Outcome

The expected outcome is a working prototype of a centralized information management system. It should make record handling more systematic, reduce dependence on scattered files, and provide quicker access to information. The project will also help in understanding practical software development, database design, authentication, API integration, and testing. The final features may be refined according to the requirements and guidance provided by the organization.

7. Future Scope

The system can later be enhanced with document upload, email notifications, dashboard analytics, approval workflows, audit logs, and integration with existing organizational software. Additional modules can be added after studying the actual working requirements of the concerned department.

Note: This is a proposed project synopsis. The exact problem, modules, and implementation plan may be modified after discussion with the concerned department during the internship.